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FISCAL INEQUITY AND TAX EVASION: AN EXPERIMENTAL APPROACH

MICHAEL W. SPICER* AND LEE A. BECKER*

ABSTRACT

This paper seeks to examine the relationship between fiscal inequity and tax evasion using an experimental approach. In the experiment, human subjects are faced with hypothetical tax evasion decisions and their behavior is observed. On the basis of socio-psychological research on inequity, it is hypothesized that the amount of taxes evaded will increase for victims of fiscal inequity but decrease for beneficiaries of fiscal inequity. The results of the experiment support this hypothesis.

TAXES have always been an important source of potential stress in the interface between a citizen and his government. There have however, in recent years, been signs that this stress is increasing. The widespread defeat of local property tax levies and bond issues, enactment in a number of states of constitutional limitations on taxing and spending and the current drive for a balanced-budget amendment to the federal constitution provide evidence of an erosion of what Caiden (1978) terms, "the political capacity to tax." Of particular concern, to tax policy-makers and administrators are signs that this stress may now be manifesting itself in the form of tax evasion.

While American taxpayers have traditionally been credited as being among the most honest in the world, recent survey research studies suggest a low level of ethics in tax compliance.¹ Also, the growth of currency relative to demand deposits since World War II leads Guttman (1977) to infer the existence of a growing "subterranean economy" of tax evaders. Feige (1979) estimates that in 1976 over \$200 billion of cash income escaped income taxation although Guttman feels this estimate may be somewhat too high (1979). In any case, tax evasion would

appear to be a significant and growing problem for tax policy-makers and administrators.

The purpose of this paper is to examine the relationship between tax evasion and perceived inequities in tax systems. While a number of survey research studies have reported positive correlations between perceptions of fiscal inequity and tax evasion,² such correlations do not allow one to infer unambiguously the causal relationship between inequity and tax evasion. Whether perceptions of inequity actually increase tax evasion or are simply a means of rationalizing illegal behavior is not clear. Our study, therefore, seeks to examine this relationship using an experimental approach. Human subjects are faced with hypothetical tax evasion decisions and their behavior is observed. This approach to studying tax evasion has been used by Friedland, Maital and Rutenberg (1978) in their examination of the effects of different probabilities of detection, levels of fines and tax rates on the tax evasion decision. While the authors do not examine empirically the relationship between fiscal inequity and tax evasion, they do suggest that an experimental approach might be useful for such an examination. This suggestion is lent support by experimental research on compliance with rules (Thibaut, Friedland and Kelley, 1974) which indicates that "equitable sharing heightens the tendency to comply."

Theoretical Background

Before discussing our experiment, it may be useful to examine some of the theoretical underpinnings of the link between tax evasion and equity. The taxpayer-government relationship is viewed here as an exchange relationship. The taxpayer foregoes a portion of his purchasing power in the private market in return for government benefits, includ-

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ing goods and services and also perhaps for non-material sources of satisfaction, such as a sense of belonging or affiliation. Research in social psychology suggests strongly that an important determinant of individual satisfaction with an exchange relationship is perceived parity or equity in the terms of trade among the participant involved in the exchange.³ For a participant, a lack of equity between his own terms of trade and those of others creates a sense of distress. This distress is felt regardless of whether the participant is the victim or the beneficiary of inequity. According to Homans (1961), if inequity is to a participant's disadvantage, he will display anger, whereas if inequity is to his advantage, then he will experience guilt feelings.

Empirical evidence also indicates that participants, in an inequitable relationship, may seek to reduce inequity by adjusting their inputs or contributions to the exchange relationship (Adams, 1965). Extending this theory to the taxpayer-government relationship, tax evasion may be seen partly as a means by which taxpayers attempt to restore equity in their terms of trade with government. In particular, it is hypothesized that if a taxpayer perceives inequity in his terms of trade with government, he will attempt to restore equity by adjusting the amount of taxes he evades. If he perceives the inequity to be to his disadvantage, he will increase the amount of taxes evaded. If he perceives the inequity to be to his advantage, then he will reduce the amount of taxes evaded. This latter proposition may seem surprising, but is consistent with the theory developed here and with empirical research in other areas of human behavior.⁴ This research indicates that not only do beneficiaries from inequity in exchange experience guilt feelings but they also attempt to assuage those feelings by increasing their contributions to the exchange. The proposition also would appear to provide a possible explanation for the inverse association between tax evasion and income reported in the study conducted by Spicer and Lundstedt (1976). Since perceptions of inequity were found in this study to stem

from the existence of unequal opportunities for tax avoidance, and since higher-income taxpayers are typically favored with greater opportunities for tax avoidance, our theory might lead us to expect less evasion from this group.

It should be noted that we are not asserting here that taxpayers are able to assess the precise value of what they and others pay to and receive from government. We simply suggest that taxpayers hold general perceptions and attitudes concerning the equity of their own and others' exchange relationships with government and that these perceptions and attitudes are relevant to the tax evasion decision.

We also wish to emphasize that the focus of the foregoing analysis is on the link between tax evasion and inequity. Other factors, such as the probabilities of detection and the level of fines, which may also affect tax evasion, are not discussed above. Our analysis, therefore, is intended to complement rather than substitute for existing economic models of tax evasion. These models are based on a view of tax evasion as gambling for illicit income, given certain probabilities of detection and fines.⁵ What is suggested in our analysis, using economic terminology, is that taxpayers' utility functions are interdependent so that the utility derived from taxes evaded depends on the taxpayer's sense of equity regarding his relationship with government. In particular, where a taxpayer perceives himself as a victim of inequity, his anger increases the marginal utility which he derives from an extra dollar of tax evasion income and hence increases the amount of taxes evaded. On the other hand, if the perceived inequity is to his advantage, guilt feelings reduce his marginal utility from tax evasion and hence the amount of taxes evaded.

Methodology

Fifty-seven University of Colorado at Colorado Springs students were recruited as participants for the experiment. Their median age was 21 to 29 years. Twenty-one participants were male and 36 were

female. The procedures closely followed the simulation paradigm reported by Friedland, Maital and Rutenberg (1978). Participants were told that the research would take the form of a tax game or simulation which would last for ten pay periods ("months"). Each "month," participants would receive a salary (the salary was the same for each participant). Each participant would then decide how much of this he/she would list as reported income, and would pay tax on that amount. Each "month," random audits would be conducted and fines imposed where taxes paid were less than taxes due on the salary amount. Participants were informed that: there was a one in 15 chance of being audited; if they were audited, they would be fined an amount equal to 15 times the tax evaded; and that the audit would be limited to the tax paid that month. At the end of 10 "months," participants were to calculate total after-tax or net income and subtract any fines. A small money prize would then be distributed to each participant according to his/her net income minus fines.

Participants received folders, each containing a tax report form, a tax table, a questionnaire and a note of additional information. The purpose of the note of information was to stimulate perceptions of inequity regarding relative tax rates. In this note, all participants were told that their own tax tables were based on a tax rate of 40 percent. However, 19 participants were told that the average tax rate was 65 percent, another 19 were told that the average tax rate was 15 percent and the rest were told the truth; i.e., that all participants had the same tax rate. This small deception was necessary because real differences in tax rates may exert an influence on tax evasion, independently of any perceptions of inequity.⁶

After participants had finished reading the sheet of additional information, the tax game commenced. A "monthly" cycle of salary announcements, tax reporting and audits was repeated 10 times. Monthly salaries were varied in order to retain participant interest. Audits were conducted in such a manner that only the par-

ticipant audited knew the results of his/her audit. The experiment lasted approximately 45 minutes.

After completing the 10 rounds and calculating their net income minus fines, participants were asked to complete the questionnaire contained in their folders. The questionnaire was used to obtain background information on participants and also their perceptions and attitudes regarding tax evasion.⁷ Included in the questionnaire was a "tax resistance" scale; an attitude scale which measures orientations toward tax evasion.⁸ The purpose of including this scale was to determine the relationship between tax evasion in an experimental setting and real world tax evasion. Attitude scales, such as the "tax resistance" scale, have typically been used to measure tax evasion in past survey research on tax evasion.⁹ This is because of the obvious difficulty in obtaining honest responses from respondents concerning their own tax evasion behavior.

Results

Participants on average evaded 23.13 percent of total taxes payable. A breakdown of this average by perceived relative tax rates lends support to our hypothesis regarding inequity and tax evasion (see Table 1). The percentage of taxes evaded was highest among those who were told that their tax rates were higher than average and lowest among those told their tax rates were lower than average.

In order to assess the significance of this result, the percentage of taxes evaded was regressed on perceived relative tax

TABLE 1
PERCENTAGE OF TAX EVADED BY
PERCEIVED RELATIVE TAX RATES

	% of Tax Evaded
"Low-Tax" Group ^a	12.26
"Medium-Tax" Group ^a	24.50
"High-Tax" Group ^a	32.63
All Groups	23.13

^aThe "Low-Tax" group includes those participants who were told that their tax rates were lower than average, the "High-Tax" group includes those told their tax rates were higher than average and the "Medium-Tax" group includes the remaining participants.

TABLE 2
REGRESSION RESULTS
(t-values in brackets)

Dependent variable = percentage of taxes evaded	Beta (standardized regression) Coefficients
Perceived relative tax rates	0.260 ^c (2.18)
Gender ^b	0.354 ^d (2.81)
Mean Tax Resistance Scores	0.189 (1.49)
Age	-0.045 (0.31)
Income	0.139 (0.95)

$R^2 = 0.288^d$

^a"Low-Tax" Group = 25%

^b"Medium-Tax" Group = 0%

^c"High-Tax" Group = +25%

^dFemale = 1

Male = 2

^eSignificant at a 95% confidence level

^fSignificant at a 99% confidence level

rates (own tax rate minus perceived average tax rate), gender, age, income and tax resistance scores. As Table 2 indicates, the Beta coefficient for perceived relative tax rates was positive and significant at a 95 percent confidence level. This result supports our hypothesis. Gender also appears to have a significant impact on tax evasion (Beta = .354). Holding other variables constant, males evade a greater percentage of their taxes than females; a result also obtained by Friedland, Maital and Rutenberg (1978). Age, income and tax resistance scores did not appear to be significantly related to tax evasion in our regression.

It is clear that these results must be interpreted with caution. The multiple regression coefficients obtained while significant were small. Furthermore, when a multivariate analysis of variance test was used, the relationship between perceived relative tax rates and taxes evaded was found to be significant only at a 90 percent confidence level.¹⁰ Nonetheless, the observed link between fiscal inequity and tax evasion is broadly consistent with the survey research findings cited earlier and should, in our view, be taken seriously.

Conclusion

The purpose of this paper has been to examine the linkage between perceptions of fiscal inequity and tax evasion using an experimental approach. Drawing on inequity theory, we have attempted to shed light on the theoretical nature of this linkage and to derive a testable hypothesis. According to this hypothesis, the amount of taxes evaded increases for victims of fiscal inequity but decreases for beneficiaries of fiscal inequity. In order to test this hypothesis, we carried out a simulation or tax game, using human subjects. Inequity was simulated by providing some participants with false information regarding relative tax rates. The results of the simulation support our hypothesis.

If our hypothesis is correct, then certain implications for tax policy follow. Attempts to redistribute tax burdens among taxpayers seem likely to impact on tax evasion. In particular, tax evasion may increase among those who feel victimized by such redistribution but decrease among those who feel favored. This being the case, the real redistributive impact of tax policies is likely to be blunted. The overall impact of such policies on tax revenues collected is not clear, but it would be a remarkable coincidence if increased revenues caused by improved compliance among some groups of taxpayers were exactly offset by reduced revenues caused by reduced compliance among other groups. Some impact on tax revenues seems likely.

It must be cautioned, however, that data on the relationship between fiscal inequity and tax evasion is still relatively sparse. More attention needs to be directed toward what forms of inequity are likely to affect tax evasion behavior. Also, it is necessary to identify more clearly the victims and beneficiaries of such inequity and their response to these forms of inequity.

Apart from any implications for fiscal inequity, the outcome of our simulation indicates that the experimental approach can play a role in understanding the factors underlying tax evasion. While the experimental approach obviously carries

with it the danger of artificiality, it enables a clearer interpretation of causal relationships than is possible using a survey approach. It is our view that neither approach alone is sufficient for the study of tax evasion. Rather, it seems more likely that useful insights into tax evasion will be obtained by using both approaches.

FOOTNOTES

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¹In a survey conducted by Spicer (1974), one-third of respondents admitted that they took deductions, which they felt were not justified or failed to report parts of their income. Furthermore, attitudes favorable to tax evasion were found to be more widely held by younger taxpayers. Song and Yarbrough (1978), in their survey study, report that American tax ethics are "barely passing." In terms of seriousness, tax evasion was ranked by respondents as only slightly more serious than stealing a bicycle.

²See Strumpel (1968), Spicer and Lundstedt (1976), Song and Yarbrough (1978).

³For a recent survey of this literature, see Walster, Walster and Berscheid (1978).

⁴See, for example, Adams and Rosenbaum (1962).

⁵See, for example, Allingham and Sandmo (1972) and Srinivasan (1973).

⁶There is some dispute as to the impact of tax rates on tax evasion. In the mathematical model presented by Allingham and Sandmo (1972), the effect of tax rates on tax evasion is found to be indeterminate. However, Yitzhaki (1974) indicates that if fines are levied on the evaded tax rather than undeclared income, as assumed by Allingham and Sandmo, then higher tax rates theoretically should induce greater income declaration. According to Pencavel (1979), Yitzhaki's conclusion may not hold if higher tax rates reduce hours worked and thereby reduce taxable income. Also, the experimental research of Friedland, Maital and Rutenberg (1978) indicates that higher tax rates reduce rather than increase the fraction of income reported.

⁷Completing the questionnaire after the experiment obviously runs the risk that questionnaire responses may be biased by the experiment. While we fully acknowledge this problem, it is our view that had we asked respondents to complete the questionnaire prior to the experiment, then the questionnaire itself might have biased responses in the experiment; a possibility we wished to avoid.

⁸Copies of the "tax resistance" scale are available from the authors on request. The scale contains 15 items derived from an inventory of 45 items using a factor analysis of responses (Spicer, 1974). The items take the form of statements regarding tax evasion and respondents are asked to indicate how well each statement fits their own feelings, experiences and behavior. Two such statements are: "When it comes to paying taxes, I say take the Internal Revenue Service for what you can get," and "I would never pad my deductions." Several of the items relate to

the respondents' own behavior and the scale appears to measure a "relative propensity" to evade taxes.

⁹Strumpel (1968) measures tax evasion in terms of attitudes towards specific acts of tax evasion committed by others. Song and Yarbrough (1978) rely on attitudes towards ethical statements regarding tax evasion.

¹⁰The analysis of variance, suggested by the reviewer, analyzed the effects of perceived relative tax rates and gender as factors, and age, income and tax resistance scores as covariates. Analysis of a variance and regression are, of course, similar techniques, but the differences between the two are sufficient to produce the small discrepancy in probability levels.

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